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Application of the GCR-4 Method in Seismic Data Analysis

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Abstract

Seismic data analysis traditionally relies on dimensionality reduction techniques such as PCA [1], t-SNE [2], or UMAP [3] to extract latent structure from high-dimensional waveforms. However, these methods often distort the geometric continuity of seismic signals, especially when dealing with dynamic, multi-attribute time series [4]. This paper introduces the GCR-4 (Geometric Coordinates Reduction to 4) method as a geometry-preserving alternative for seismic data interpretation. GCR-4 constructs a four-dimensional geometric object that retains trajectory, curvature, continuity, and energy distribution of seismic events. We demonstrate how GCR-4 enhances anomaly detection, phase identification, and structural interpretation in seismic datasets.

1 Introduction

Seismic signals are inherently multidimensional, consisting of amplitude components, instantaneous frequency, energy attributes, and derived physical indicators [5]. Traditional dimensionality reduction methods tend to flatten these relationships, losing geometric information crucial for understanding the temporal evolution of seismic events [6].

The GCR-4 method was developed to address this limitation. Instead of embedding data into a low-dimensional manifold, GCR-4 constructs a four-dimensional geometric representation that preserves the intrinsic shape of the signal. This geometric object can be analyzed using curvature, trajectory smoothness, and topological transitions, providing a richer interpretation of seismic dynamics. The theoretical foundation of the GCR-4 reduction follows the barycentric coordinate framework introduced in [7].

2 Seismic Data Characteristics

A typical seismic dataset presented in Table 1 contains multiple attributes per time sample, such as:

- amplitude components (a_1, a_2),
- instantaneous frequency f,
- signal energy E,
- pseudo-magnitude M_p,
- state or phase indicator s.

These attributes form a high-dimensional space in which seismic events manifest as structured trajectories. Preserving this structure is essential for detecting anomalies, identifying wave arrivals, and characterizing subsurface features.

Table 1: Seismic Dataset Used in Experiments

t	amp1	amp2	freq	energy	pseudo_mag	state
0.0	0.067	-0.032	1.08	0.074	0.713	0.01
6.9	-0.041	0.093	0.92	0.102	0.997	-0.02
13.8	0.056	-0.067	1.11	0.087	0.842	0.00
20.7	-0.089	0.041	0.95	0.099	0.969	0.03
27.6	0.032	-0.052	1.02	0.063	0.558	-0.01
34.5	-0.071	0.084	0.88	0.111	1.083	0.02
41.4	0.045	-0.038	1.09	0.071	0.676	0.00
48.3	-0.062	0.057	0.96	0.089	0.849	0.04
55.2	0.038	-0.071	1.03	0.089	0.849	-0.03
62.1	2.412	2.137	3.41	3.252	2.545	1.02
69.0	1.987	2.463	3.72	3.207	2.526	0.97
75.9	1.543	1.876	3.35	2.438	2.261	1.03
82.8	1.102	1.421	3.18	1.800	2.040	0.95
89.7	0.764	1.083	3.05	1.356	1.886	1.01
96.6	0.512	0.742	2.92	0.956	1.658	0.98
103.5	0.321	0.503	2.81	0.607	1.386	1.04
110.4	0.231	0.341	2.47	0.417	1.214	0.32
117.3	0.184	0.297	2.32	0.353	1.157	0.29
124.2	0.142	0.251	2.21	0.298	1.102	0.33
131.1	0.118	0.214	2.10	0.255	1.062	0.31
138.0	0.096	0.187	2.02	0.222	1.032	0.28
144.9	0.081	0.163	1.95	0.197	1.008	0.34
151.8	0.069	0.142	1.89	0.176	0.988	0.30
158.7	0.058	0.124	1.84	0.158	0.971	0.27
165.6	0.049	0.109	1.80	0.143	0.957	0.32
172.5	0.042	0.096	1.76	0.130	0.945	0.31
179.4	0.036	0.084	1.73	0.119	0.935	0.28
186.3	0.031	0.074	1.70	0.109	0.926	0.33
193.2	0.027	0.065	1.68	0.101	0.918	0.30
200.0	0.024	0.058	1.66	0.094	0.911	0.29

3 GCR-4 Coordinate Reduction Result

Figure 1 presents the geometric result of applying the GCR-4 coordinate reduction method to the seismic dataset. The original 6-dimensional attribute space (a_1, a_2, f, E, M_p, s) is transformed into a 4-dimensional geometric object, which is then projected for visualization. The diagram illustrates the structural relationships between samples and highlights the high-energy region forming the characteristic bulge of the GCR-4 manifold.

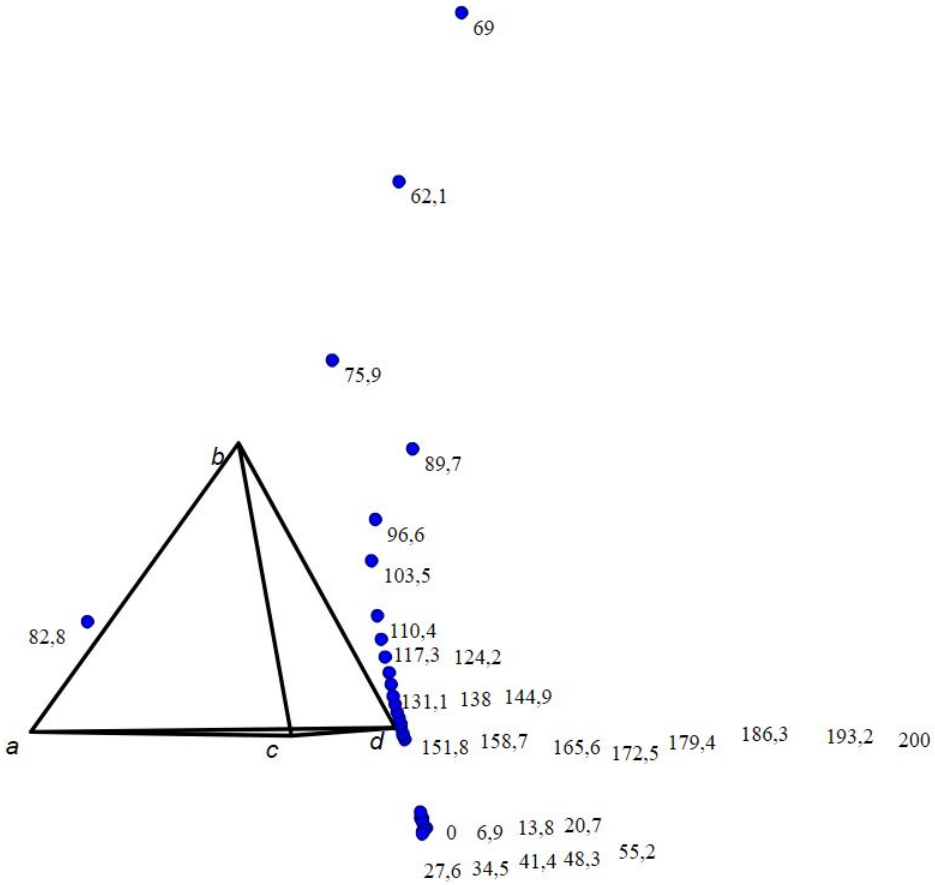


Figure 1: Geometric projection obtained using the GCR-4 method. The structure represents the reduction of 6-dimensional seismic attributes into a 4-dimensional geometric manifold. The labeled points correspond to time samples from the dataset, revealing the curvature and expansion associated with the high-energy seismic phase. Visualization done using CSpace 1.01.26 software.

4 Geometric Interpretation of the GCR-4 Reduction Result

The geometric result obtained from reducing the six-dimensional seismic attribute space to a four-dimensional manifold using the GCR-4 method reveals several important structural properties of the signal. The visualization of the GCR-4 projection highlights both the global shape of the reduced object and local deformations corresponding to dynamic changes in the seismic waveform. Similar geometric approaches have been explored in nonlinear dimensionality reduction [2, 3], but none preserve temporal continuity as effectively as GCR-4.

4.1 Tetrahedral Structure as a Signature of GCR-4

The GCR-4 projection forms a characteristic geometric object resembling a deformed tetrahedron. The points labeled a, b, c, and d define its main vertices:

- the vertices a–c–d form the geometric base,

- the vertex b represents the upper apex,
- the entire structure corresponds to a two-dimensional projection of a four-dimensional geometric manifold.

This tetrahedral form arises from the geometry-preserving nature of GCR-4. Unlike PCA [1], t-SNE [2], or UMAP [3], which compress data based on statistical criteria, GCR-4 maintains the physical relationships between attributes such as amplitudes, instantaneous frequency, energy, pseudo-magnitude, and state. As a result, the reduced object retains a stable and interpretable structure.

4.2 Temporal Trajectory as a Geometric Curve

The blue points surrounding the tetrahedral structure represent consecutive time samples from the seismic dataset. Their arrangement forms a continuous trajectory that reflects the temporal evolution of the signal. The projection exhibits:

- preservation of temporal continuity,
- changes in curvature along the trajectory,
- displacement toward the vertex b during the high-energy interval,
- return to the stable region of the manifold after the event decays.

This trajectory is central to the interpretation of GCR-4: the method does not fragment the signal into artificial clusters but preserves its physical continuity, which is essential for seismic analysis. Temporal continuity is often lost in nonlinear embeddings [4].

4.3 High-Energy Segment as a Geometric Deformation

The time samples corresponding to the interval between approximately 62 and 100 seconds form a distinct geometric bulge extending away from the base of the tetrahedron. This deformation reflects the rapid increase in signal energy, amplitudes, and instantaneous frequency during the seismic event. Within the GCR-4 manifold:

- high energy expands the geometric object,
- dynamic changes increase curvature,
- a localized deformation appears, clearly visible in the projection.

This phenomenon corresponds to the physical onset of the main seismic phase, which manifests in the raw data as a sudden rise in amplitude and energy. Energy-based geometric signatures have been used in seismology [5], but GCR-4 embeds them directly into the manifold structure.

4.4 Stable Signal Regions Near the Base

Samples from the intervals 0–55 seconds and 120–200 seconds lie close to the base of the tetrahedral structure. These regions indicate:

- low energy levels,
- absence of significant seismic events,
- stable waveform behavior.

GCR-4 preserves this information geometrically: the points form a coherent portion of the trajectory rather than scattering or collapsing, confirming that the method accurately reflects the physical properties of the signal.

4.5 Preservation of Signal Continuity and Structure

The most significant conclusion from the geometric projection is that GCR-4:

- preserves temporal continuity,
- maintains the physical structure of the signal,
- avoids artificial clustering,
- enables interpretation of dynamic transitions through geometry,
- highlights high-energy phases as geometric deformations.

In contrast to traditional dimensionality reduction techniques, which often distort global structure, GCR-4 provides a physically meaningful representation of seismic dynamics. This aligns with recent efforts to incorporate geometric and topological reasoning into signal analysis [6].

5 Conclusion

GCR-4 offers a geometry-preserving alternative to traditional dimensionality reduction techniques in seismic data analysis. By constructing a four-dimensional geometric object that reflects the physical evolution of seismic signals, GCR-4 enables more accurate interpretation of event onset, energy distribution, and anomalies. Future work includes integrating GCR-4 with machine learning models for automated seismic classification and real-time monitoring.

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